

V4.2 Toy Model 1D-C3

Inverted Radii: Wide Connectivity + Tight Radial Delay

Purpose:

Test Q Twin's field inversion correction after C2 became too wall-like.

C3 geometry:

$\rho_{\lambda 0} = 4.0$: wide minor scalar delay / general congestion.

$\rho_{\mu} = 4.0$: wide vectorial connectivity / early relational capture.

$\rho_{\lambda} = 2.0$: tight radial delay / core penalty.

Cost rule:

$$c_{ij} = c0_{ij} * [1 + \lambda_{\lambda 0} R_{\lambda 0} + \lambda_r R_{\lambda} (e_{\hat{}} \cdot r_{\hat{}})^2] / [1 + \mu R_{\mu} (e_{\hat{}} \cdot g_{\hat{}})^2]$$

Formula meaning:

$\lambda_{\lambda 0}$ = minor scalar/general delay sensitivity, fixed at 0.5 in main cases.

λ_r = tight radial directional delay sensitivity.

μ = wide vectorial connectivity sensitivity.

$(e_{\hat{}} \cdot r_{\hat{}})^2$ = radial alignment penalty for delay.

$(e_{\hat{}} \cdot g_{\hat{}})^2$ = radial alignment shortcut for connectivity.

Diagnostics:

$F_{\text{center}} > 0$ = off-axis paths moved inward near centre.

core paths = bundle paths entering the tight radius $\rho_{\lambda} = 2.0$.

Goldilocks G = rough heuristic: positive focusing but not excessive core collapse or huge detour.

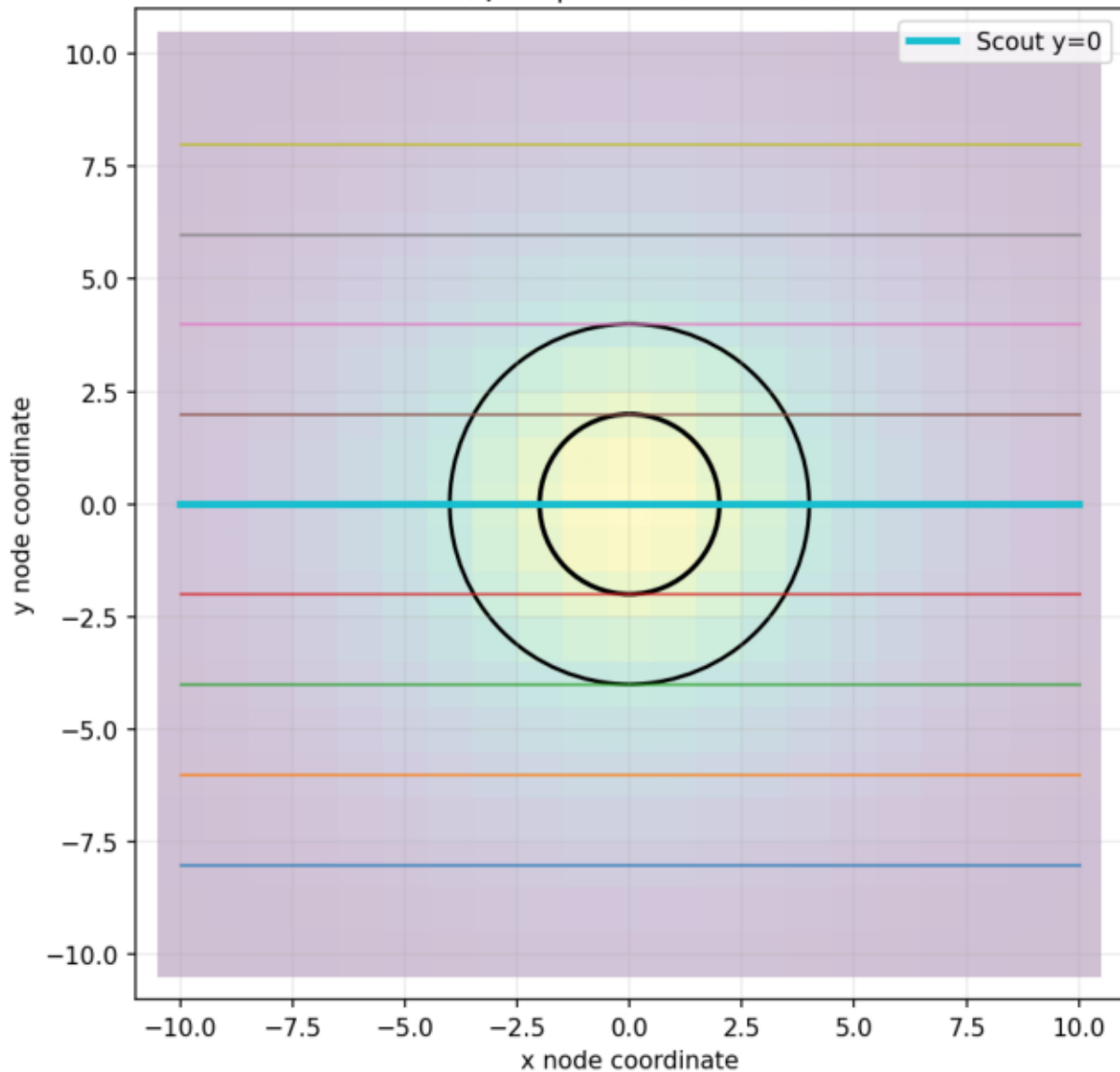
Guardrail:

This is not General Relativity and not gravitational lensing. It is a graph test of whether wide directional connectivity plus tight radial delay creates more lensing-like path behaviour.

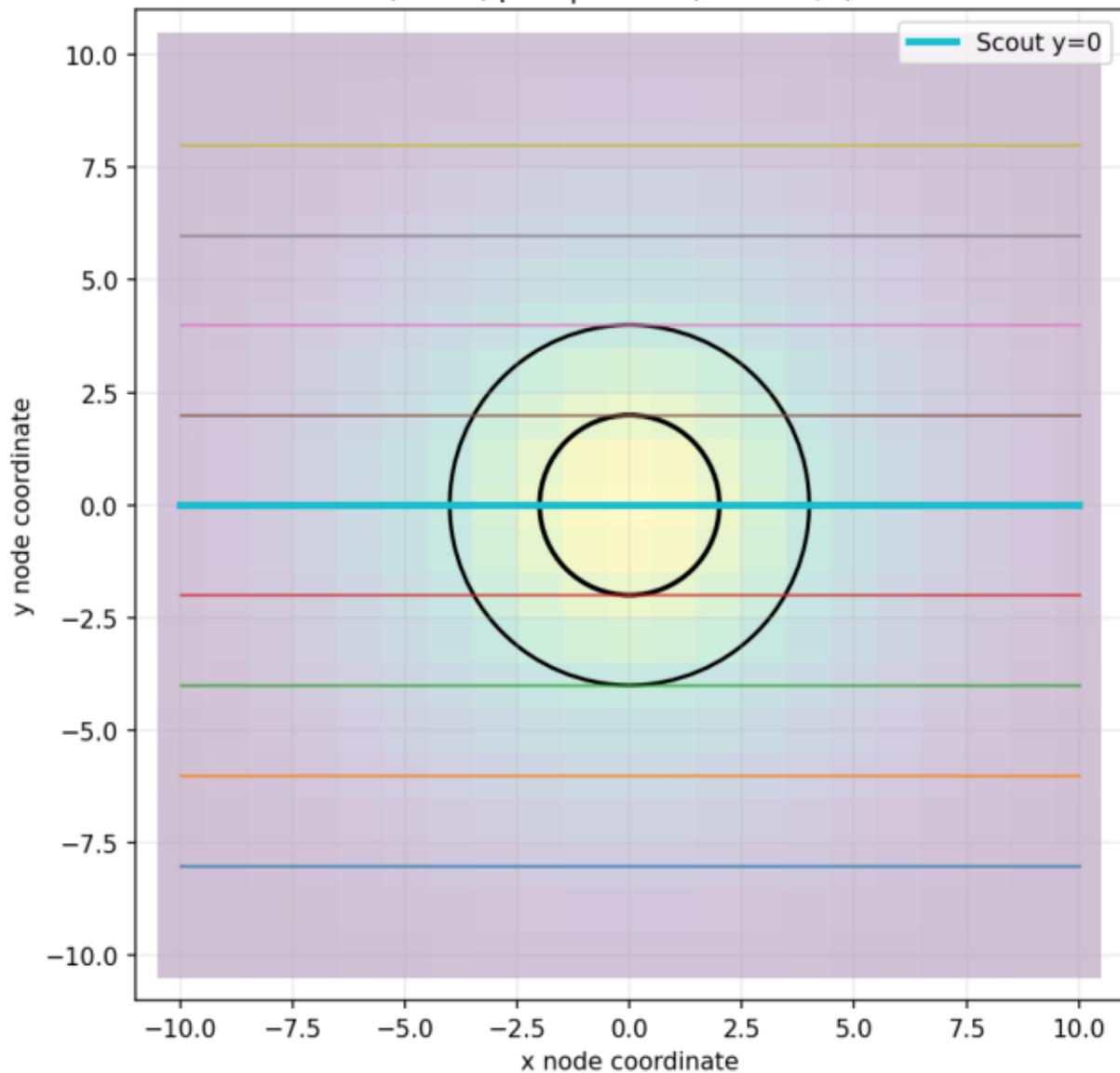
Numerical summary:

C3_flat_control: $\lambda_r = 0.0$, $\mu = 0.0$, $L = 0.00$, $\text{avgD} = 0.00$, $F = 0.00$, $\text{det} = 0.00$, $\text{wide} = 5/9$, $\text{core} = 3/9$, $G = 0.00$
C3_lambda0_only: $\lambda_r = 0.0$, $\mu = 0.0$, $L = 4.95$, $\text{avgD} = 2.70$, $F = 0.00$, $\text{det} = 0.00$, $\text{wide} = 5/9$, $\text{core} = 3/9$, $G = 0.00$
C3_connect_only_mu2: $\lambda_r = 0.0$, $\mu = 2.0$, $L = -8.70$, $\text{avgD} = -5.20$, $F = 5.00$, $\text{det} = -5.00$, $\text{wide} = 9/9$, $\text{core} = 9/9$, $G = 2.60$
C3_core_delay_only: $\lambda_r = 2.0$, $\mu = 0.0$, $L = 6.84$, $\text{avgD} = 3.07$, $F = -0.75$, $\text{det} = 0.75$, $\text{wide} = 3/9$, $\text{core} = 0/9$, $G = -0.75$
C3_lamr1_mu2: $\lambda_r = 1.0$, $\mu = 2.0$, $L = -4.71$, $\text{avgD} = -1.41$, $F = 2.25$, $\text{det} = -1.75$, $\text{wide} = 7/9$, $\text{core} = 7/9$, $G = 0.65$
C3_lamr2_mu2: $\lambda_r = 2.0$, $\mu = 2.0$, $L = -2.89$, $\text{avgD} = -0.77$, $F = 0.25$, $\text{det} = 0.00$, $\text{wide} = 5/9$, $\text{core} = 3/9$, $G = 0.25$
C3_lamr2_mu3: $\lambda_r = 2.0$, $\mu = 3.0$, $L = -5.90$, $\text{avgD} = -2.62$, $F = 3.00$, $\text{det} = -2.00$, $\text{wide} = 9/9$, $\text{core} = 7/9$, $G = 1.40$
C3_lamr3_mu3: $\lambda_r = 3.0$, $\mu = 3.0$, $L = -4.52$, $\text{avgD} = -2.04$, $F = 2.50$, $\text{det} = -1.75$, $\text{wide} = 9/9$, $\text{core} = 5/9$, $G = 1.70$
C3_lamr3_mu5: $\lambda_r = 3.0$, $\mu = 5.0$, $L = -8.41$, $\text{avgD} = -5.19$, $F = 3.75$, $\text{det} = -2.75$, $\text{wide} = 9/9$, $\text{core} = 9/9$, $G = 1.35$
C3_lamr4_mu6: $\lambda_r = 4.0$, $\mu = 6.0$, $L = -8.84$, $\text{avgD} = -5.83$, $F = 3.25$, $\text{det} = -2.25$, $\text{wide} = 9/9$, $\text{core} = 9/9$, $G = 0.85$

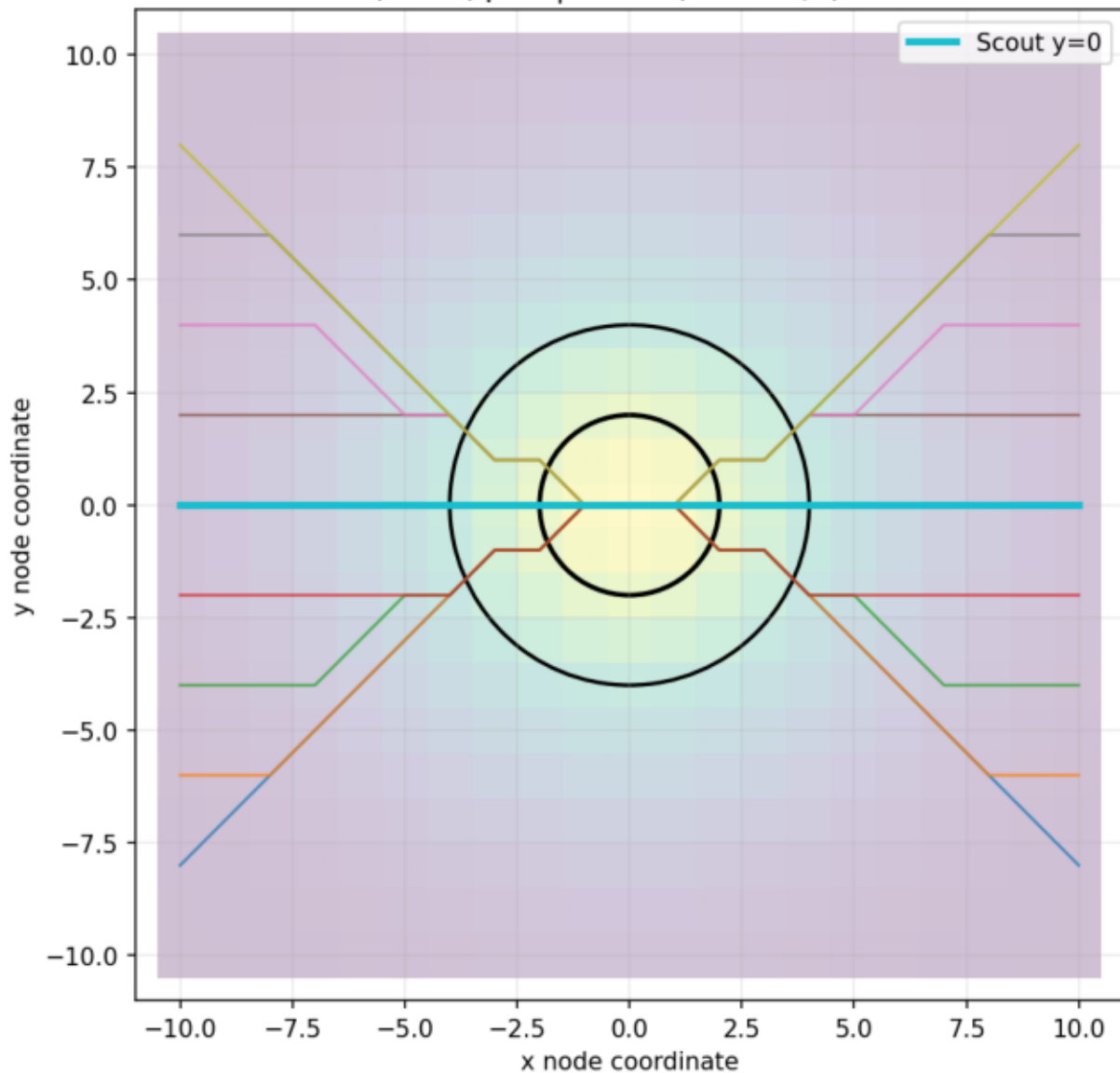
C3_flat_control
 $\lambda_0=0, \lambda_r=0, \mu=0 \mid \bar{F}=0.00, \text{core}=3/9, G=0.00$



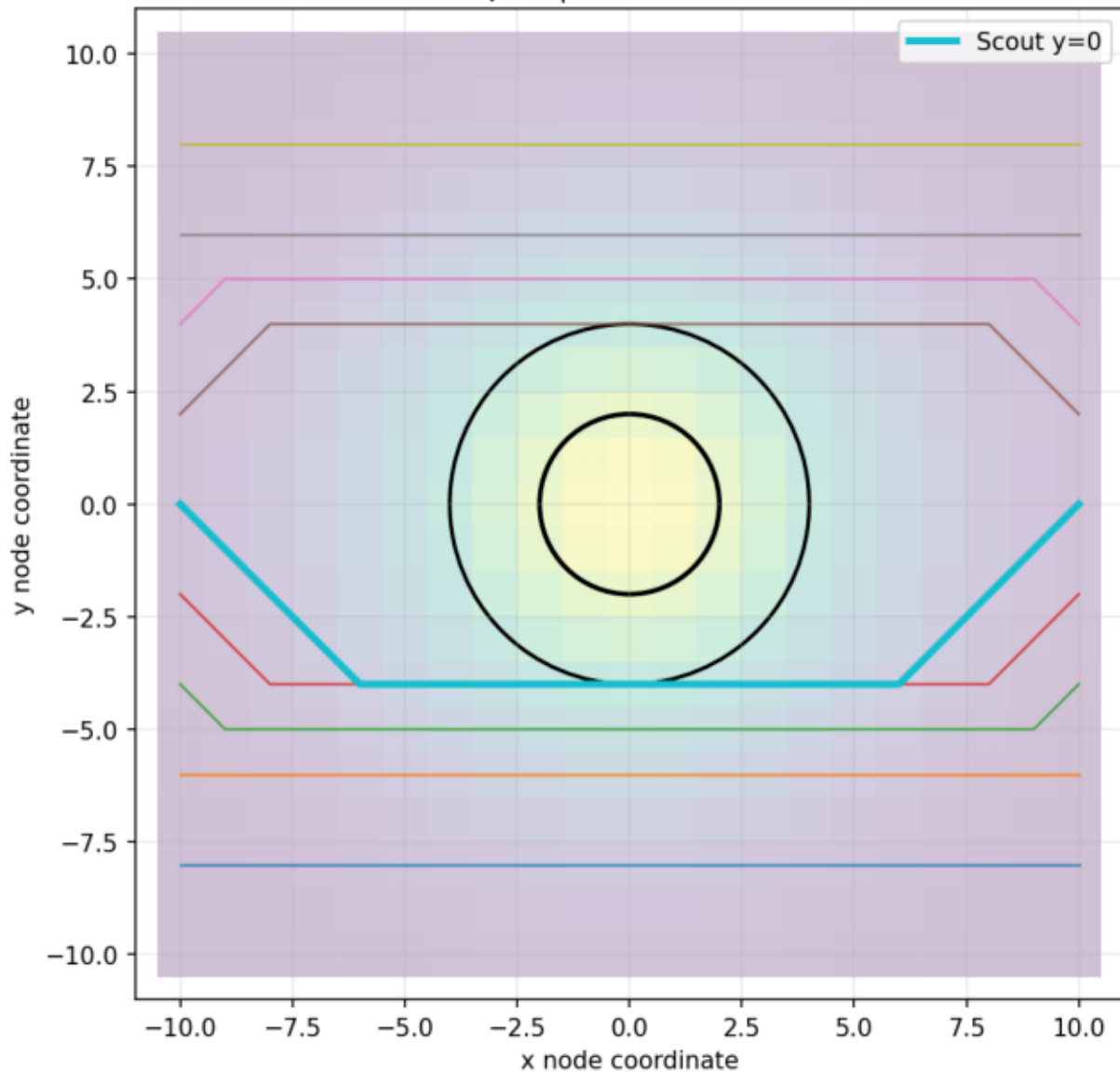
C3_lambda0_only
 $\lambda_0=0.5, \lambda_r=0, \mu=0 \mid F=0.00, \text{core}=3/9, G=0.00$



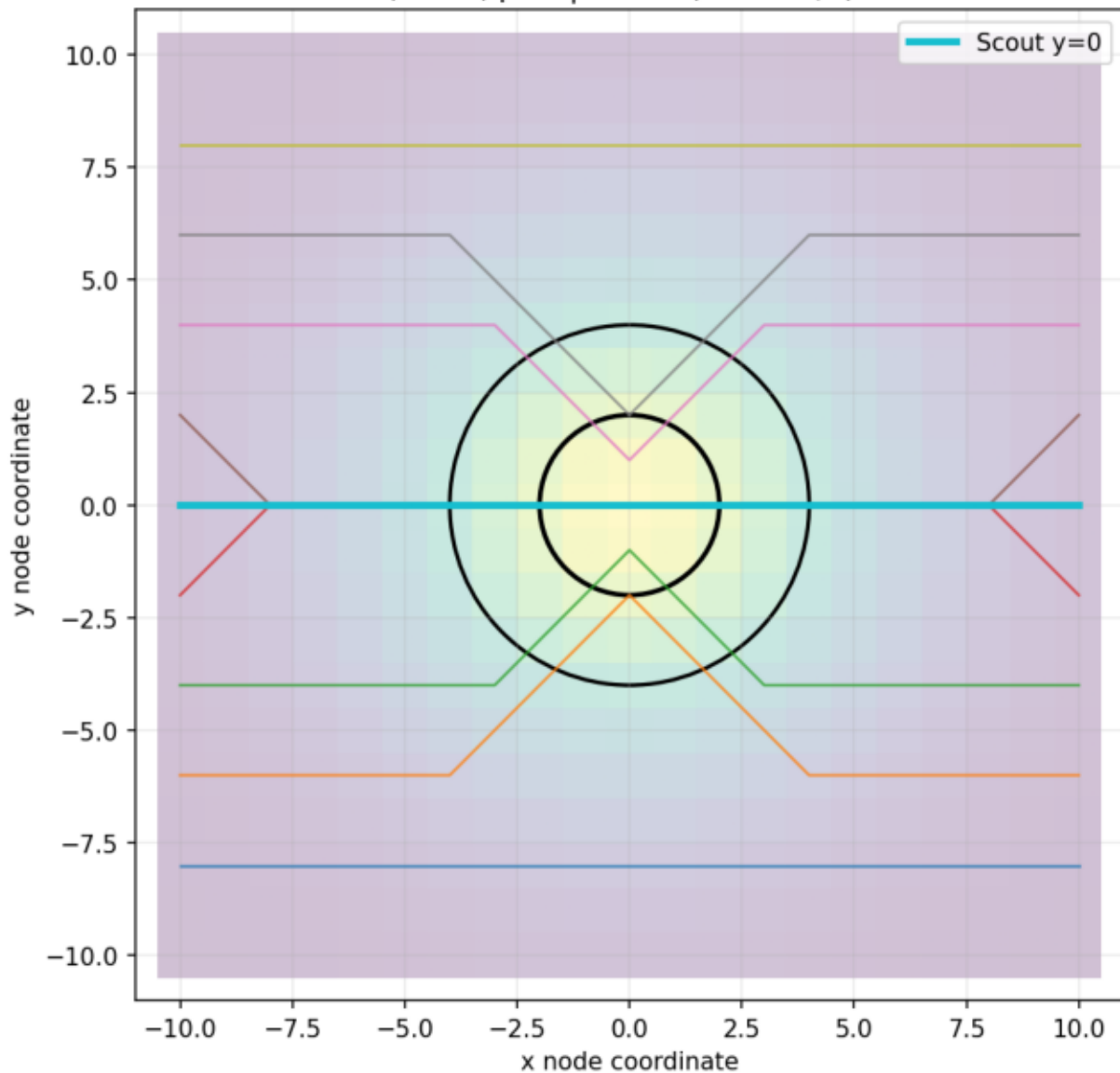
C3_connect_only_mu2
 $\lambda_0=0, \lambda_r=0, \mu=2 \mid F=5.00, \text{core}=9/9, G=2.60$



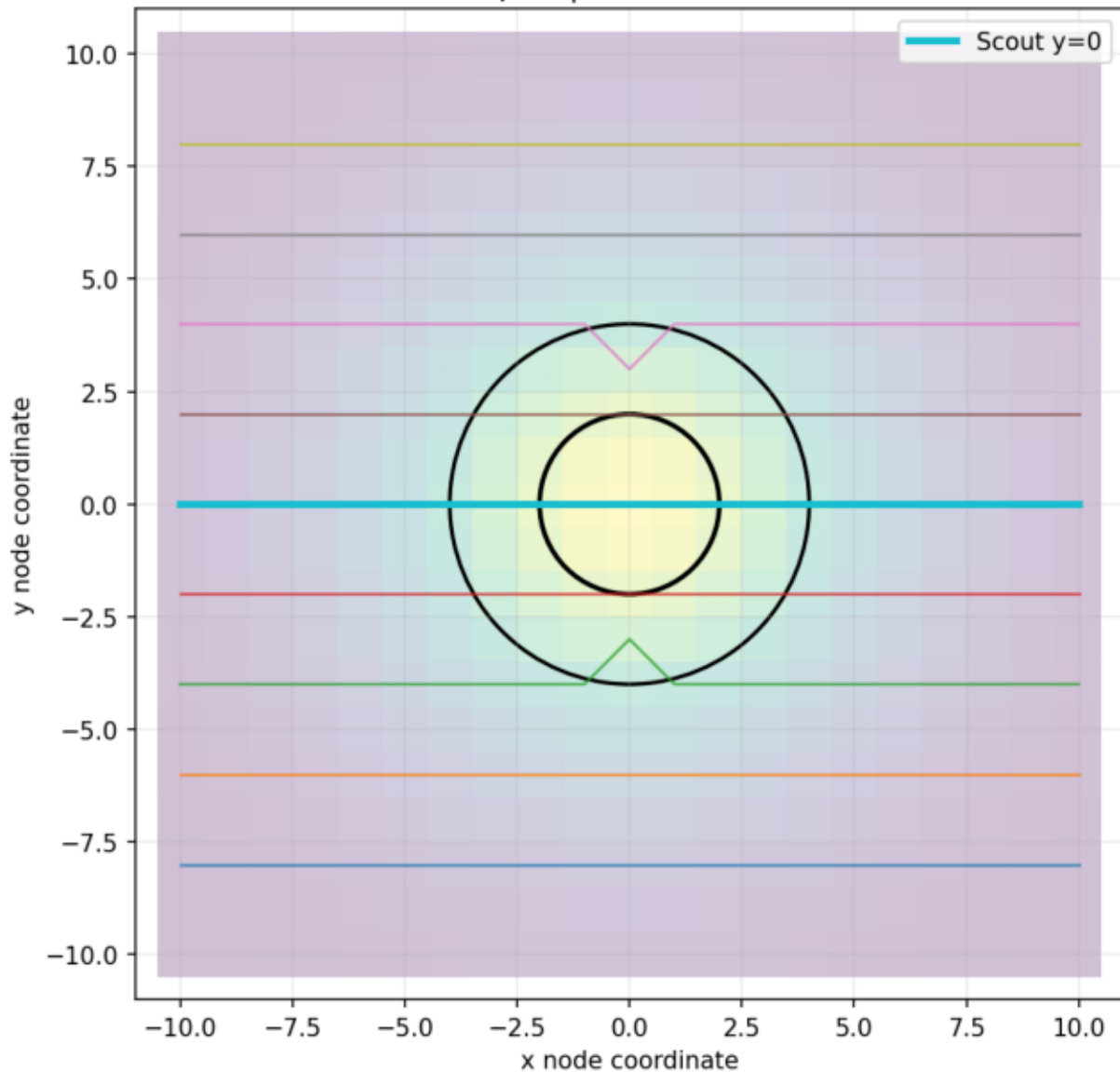
C3_core_delay_only
 $\lambda_0=0.5, \lambda_r=2, \mu=0 \mid F=-0.75, \text{core}=0/9, G=-0.75$



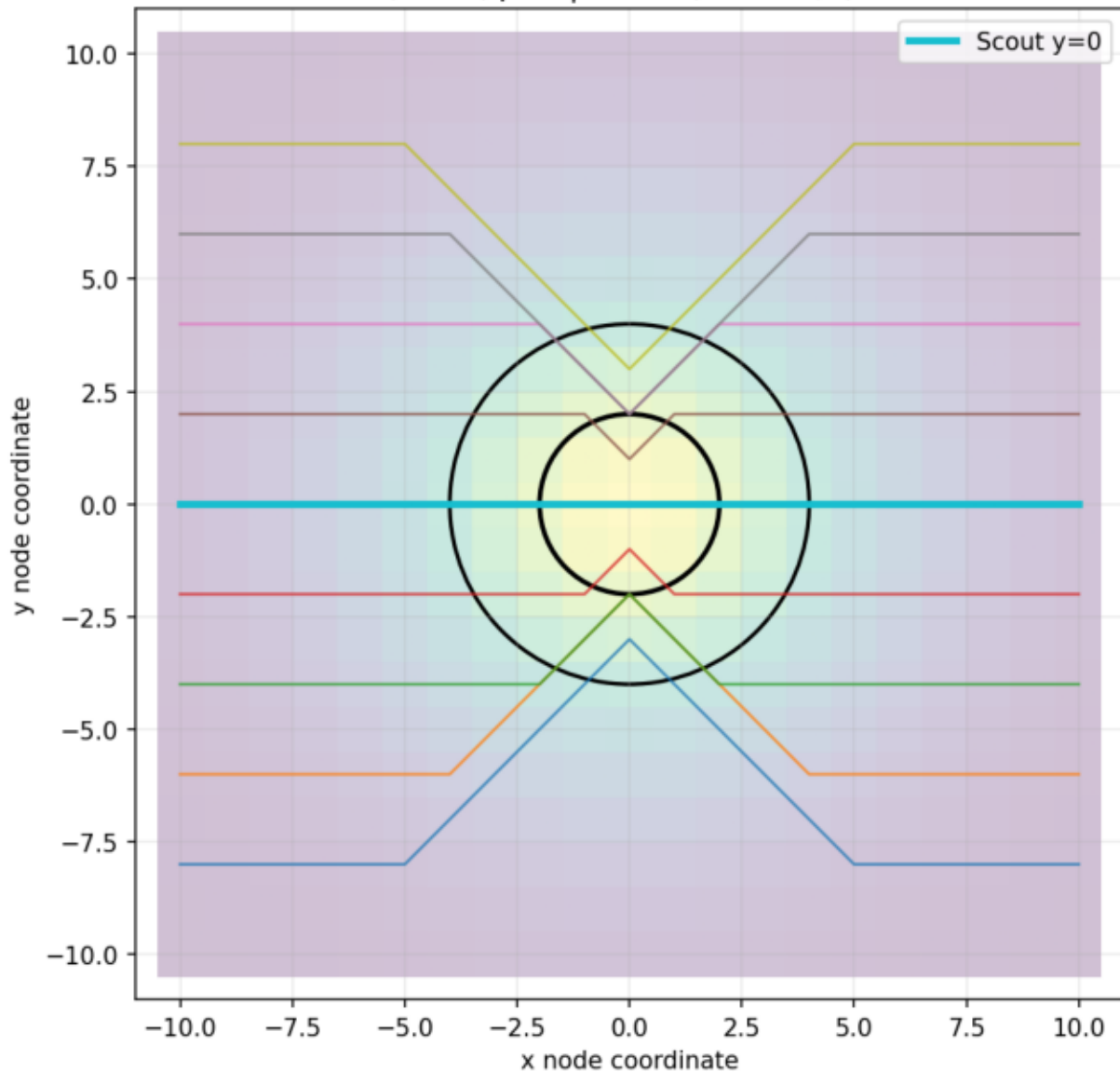
C3_lamr1_mu2
 $\lambda_0=0.5$, $\lambda_r=1$, $\mu=2$ | $F=2.25$, $\text{core}=7/9$, $G=0.65$



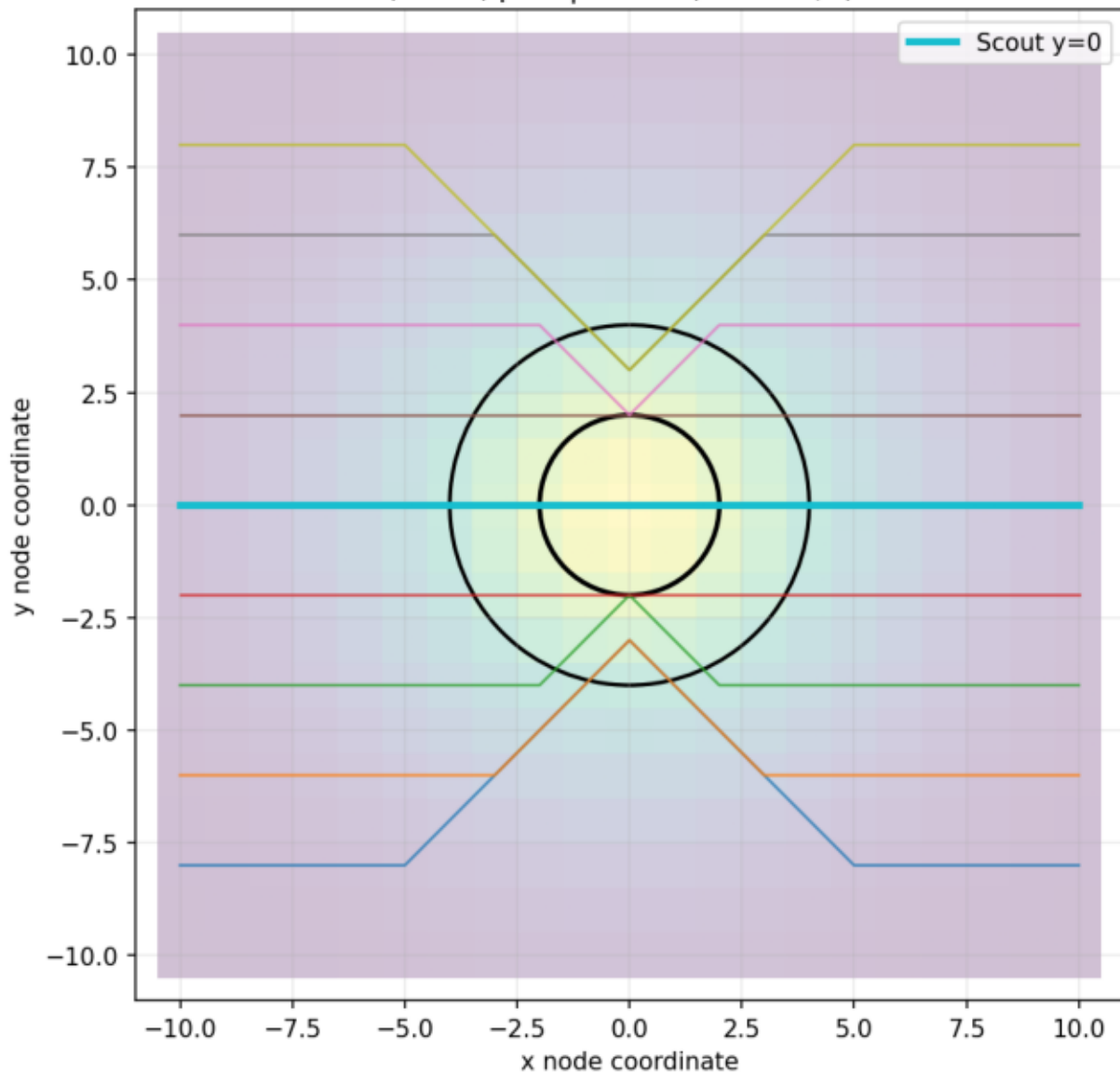
C3_lamr2_mu2
 $\lambda_0=0.5, \lambda_r=2, \mu=2 \mid F=0.25, \text{core}=3/9, G=0.25$



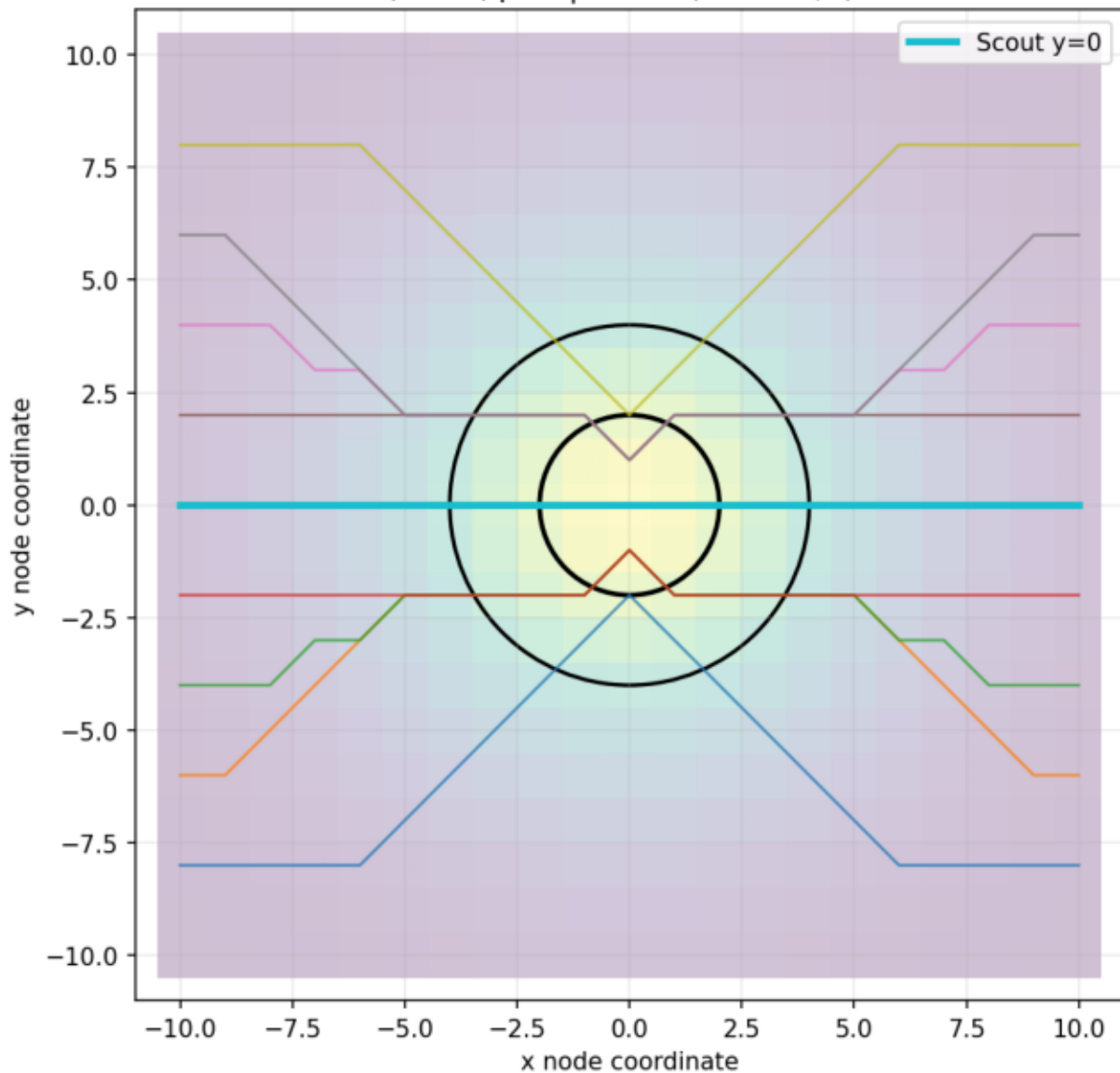
C3_lamr2_mu3
 $\lambda_0=0.5$, $\lambda_r=2$, $\mu=3$ | $F=3.00$, $\text{core}=7/9$, $G=1.40$



C3_lamr3_mu3
 $\lambda_0=0.5$, $\lambda_r=3$, $\mu=3$ | $F=2.50$, $\text{core}=5/9$, $G=1.70$



C3_lamr3_mu5
 $\lambda_0=0.5$, $\lambda_r=3$, $\mu=5$ | $F=3.75$, core=9/9, $G=1.35$



C3_lamr4_mu6
 $\lambda_0=0.5$, $\lambda_r=4$, $\mu=6$ | $F=3.25$, core=9/9, $G=0.85$

